

Connor Lane

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Education

University of California, Santa Barbara, Ph.D. in Mathematics Sept 2025 – TBD

Rose-Hulman Institute of Technology, BS in Mathematics Summa Cum Laude Sept 2022 – May 2025

- Minors in Computer Science and Theoretical Physics
- Clarence P. Sousley Award for excellence in mathematics graduates
- Senior Thesis: “Cyclotomic Units, Volkenborn Integration, and p-adic L-functions.” Advised by Dr. Tim All

Publications & Preprints

Dispersive estimates for Dirac Operators in dimension four with obstructions at threshold energies Oct 2025

Green, W.R., Lane, C. & Lyons, B. Dispersive estimates for Dirac Operators in dimension four with obstructions at threshold energies. *Calc. Var.* 64, 292 (2025). <https://doi.org/10.1007/s00526-025-03163-7>

AGM Aquariums and Elliptic Curves Over Arbitrary Finite Fields Apr 2025

June Kayath, Connor Lane, Ben Neifeld, Tianyu Ni, Hui Xue, AGM Aquariums and Elliptic Curves Over Arbitrary Finite Fields, *Res. Number Theory* 11 (2025), no. 2, Paper No. 48, 31 pp.; MR4890995

The massless Dirac equation in three dimensions: dispersive estimates and zero energy obstructions. Jan 2025

William R. Green, Connor Lane, Benjamin Lyons, Shyam Ravishankar, Aden Shaw. The massless Dirac equation in three dimensions: Dispersive estimates and zero energy obstructions, *J. Differential Equations* 416 (2025), 449–490; MR4806982

Subspaces Spanned by Eigenforms with Nonvanishing Central L-Values Jul 2024

June Kayath, Connor Lane, Ben Neifeld, Tianyu Ni, Hui Xue. "Subspaces Spanned by Eigenforms with Nonvanishing Central L-Values" arXiv preprint arXiv:2407.00532. (2024)

Asymptotic Distribution of Residues in Pascal’s Triangle mod p Sep 2023

Connor Lane. Asymptotic Distribution of Residues in Pascal’s Triangle mod p. arXiv preprint arXiv:2309.12942 (2023)

Remote Work: Fad or Future Apr 2022

Aruni Chenxi, Nika Chuzhoy, Max Hartman, Connor Lane, and Nathan Liu "Remote Work: Fad or Future." Mathworks Math Modeling Challenge Runner-Up, <https://m3challenge.siam.org/resources/archives/2022-year-at-a-glance/2022-winningsolutions/> (2022)

Talks Given

- “The Prime At Infinity and Other Lies in Algebraic Number Theory,” Ross Mathematics Program Dorm Lecture, July 24 2025
- “Math Until We Die,” Ross Mathematics Program Dorm Lecture, July 7 2025
- “How I Rigorously Broke a Painting,” Ross Mathematics Program Dorm Lecture, June 20 2025
- “AGM Aquariums and Elliptic Curves Over Finite Fields,” Rose-Hulman Undergraduate Mathematics Conference 2025, March 29 2025
- “AGM Aquariums and Elliptic Curves Over Finite Fields,” Joint Mathematics Meetings PME Undergraduate Poster Session, January 9 2025
- “AGM Aquariums and Elliptic Curves Over Finite Fields,” Clemson University Undergraduate Research Symposium, July 26 2024

- “Cyclic Galois Extensions and Groups on Conics,” Rose-Hulman Undergraduate Mathematics Conference 2024, April 27 2024
- “Asymptotic Distribution of Residues in Pascal’s Triangle mod p ,” Joint Mathematics Meetings PME Undergraduate Poster Session, January 6 2024
- “Infinite Galois Theory,” Ross Mathematics Program Dorm Lecture, July 15 2023
- “The Riemann Zeta Function and a Problem on my Shirt,” Ross Mathematics Program Dorm Lecture, July 6 2023
- “Pascal’s Triangle mod p ,” Rose-Hulman Undergraduate Mathematics Conference 2023, April 21 2023

Teaching Experience

Teaching Assistant , University of California, Santa Barbara	Oct 2025 – Ongoing
• Ran recitation sections for undergraduate classes • Made original classroom materials	
Counselor , Ross Mathematics Program	Summer 2023, 2025
• Taught and graded number theory for a “family” of 5-6 high school students • Instructor and course designer for Algebraic Number Theory	
Grader , Rose-Hulman Institute of Technology	Dec 2023 – Mar 2025
• Graded for upper level classes – Real Analysis, Algebraic Geometry, Galois Theory	
Tutor , Rose-Hulman Institute of Technology	Sep 2023 – Mar 2025
• Tutored nearly every class offered by the math department • Designed finals review packet for introduction to proofs	

Academic Honors

Clarence P. Sousley Award , Rose-Hulman Institute of Technology Math Dept.	2025
Ying Award for Best Student-Faculty Collaboration , Rose-Hulman Institute of Technology	2025
Goldwater Scholarship Winner , Goldwater Foundation	2024
Theodore Paine Palmer Award , Rose-Hulman Institute of Technology Math Dept.	2023
Putnam Top 200 , Mathematical Association of America	2023-2025
Dean’s List , Rose-Hulman Institute of Technology	2022-2025
M3 Challenge Runner-Up , Society for Industrial and Applied Math	2022

Selected Coursework

Topics in Geometry	Fall 2025
University of California, Santa Barbara: Cheeger-Gromoll splitting theorem, Cheeger-Gromoll soul theorem	
Complex Manifolds	Spring 2025
Rose-Hulman Institute of Technology: Complex analysis in several variables, Hartog’s phenomenon, geometry of complex manifolds	
Forcing	Spring 2025
Rose-Hulman Institute of Technology: Independence of CH, Whiteheads problem, applications to analysis	
Programming Language Paradigms	Spring 2025
Rose-Hulman Institute of Technology: Functional and declarative programming, uncommon language features	
Topics in Algebraic Geometry	Winter-Spring 2025
Rose-Hulman Institute of Technology: Sheaf cohomology on sites, étale cohomology of schemes, Poincare duality on varieties, applications to algebraic number theory	

Advanced Elliptic Curves

Fall 2024

Rose-Hulman Institute of Technology: Integral points, non-triviality of the Tate-Shafarovich group, elliptic surfaces, Néron models

General Relativity

Spring 2024

Rose-Hulman Institute of Technology: Einstein's field equation, black holes, applications to cosmology

Cohomology of Number Fields

Winter 2024

Rose-Hulman Institute of Technology: The local-to-global principle, Grunwald-Wang, genus theory

Differential Forms in Algebraic Topology

Winter 2024

Rose-Hulman Institute of Technology: De Rham cohomology, Characteristic classes, spectral sequences